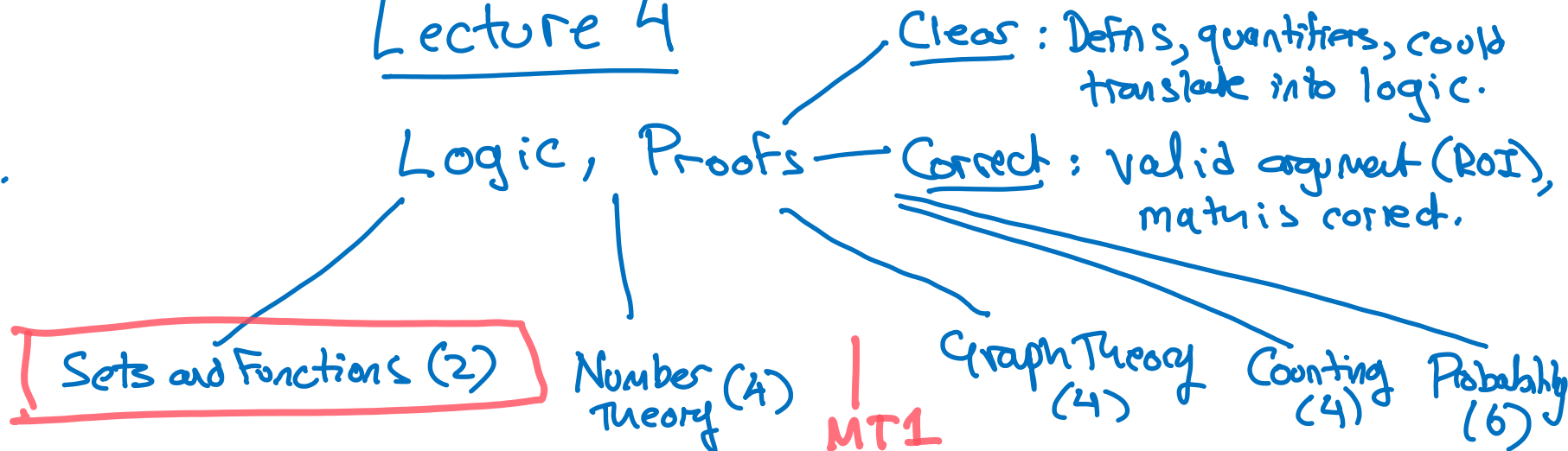


Lecture 4

So far:



"will build from scratch."

5% HW
2 drops

> 13% Quizzes
2 drops

80% exams.
lowest MT replaces final.

I Sets

II Functions

Defn: A set is an unordered collection of objects called its elements. A set contains its elements.

Denoted $x \in A$ "x is an element of A."

$x \notin A$ "... not ..."

Key point: For every x , " $x \in A$ " is a proposition.

3 ways to specify : ① List: $A = \{1, 2, 3\}$, $C = \{Nikhil, \dots\}$
for finite sets.

② As a 'truth set': $\mathbb{Z} = \{x : x \text{ is an integer}\}$.
in general, given $P(x)$ $A = \{x : P(x)\}$ is a set.
property $\swarrow$
 $C = \{x : x \text{ is a person in this class}\}$
 $\mathbb{R} = \{x : x \text{ is a real number}\}$
 $\mathbb{Q} = \{ \underbrace{x \in \mathbb{R}}_{\text{rational}(x)} : \exists p, q \in \mathbb{Z} (q \neq 0 \wedge x = p/q) \}$

③ Implicit: $\mathbb{Z} = \{ \dots, -2, -1, 0, 1, 2, \dots \}$

Defn: The set with no elements is called the empty set,
denoted $\emptyset$. Formally, $\forall x (x \neq \emptyset)$.

Q) Can an element appear more than once?

A) $\{1, 2, 2\}$ multiset, for now equivalent to $\{1, 2\}$.

Defn (Equality) Two sets $A; B$ are equal if they contain the same elements.

$$\text{formally } A=B \iff \forall x (x \in A \iff x \in B)$$

Defn: (Subset) A is a subset of B if every element of A is an element of B .

$$A \subseteq B \iff \forall x (x \in A \rightarrow x \in B).$$

ex: $\mathbb{Q} \subseteq \mathbb{R}. \quad \forall x (x \in \mathbb{Q} \rightarrow x \in \mathbb{R}).$

$$\mathbb{R} \not\subseteq \mathbb{Q} \quad \neg \forall x (x \in \mathbb{R} \rightarrow x \in \mathbb{Q})$$

$$\exists x (x \in \mathbb{R} \wedge x \notin \mathbb{Q})$$

Pf: Consider $\sqrt{2} \in \mathbb{R}$,
 $\sqrt{2} \notin \mathbb{Q}$ by Lec 3. $\square$

(Q) If $A=B$ is $A \subseteq B$? (A) $A=B \rightarrow A \subseteq B$.

Prop 1: If A, B, C are sets and $A \subseteq B$ and $B \subseteq C$, then
 $A \subseteq C$.

$\forall A, B, C$ set $(A \subseteq B \wedge B \subseteq C \rightarrow A \subseteq C)$.

Proof: Assume A, B, C are arbitrary. $\forall$

Assume $A \subseteq B$ and $B \subseteq C$. premises

We will show $A \subseteq C$.

Let x be an arbitrary element of A . $\leftarrow$

Since $A \subseteq B$, $x \in B$.

Since $B \subseteq C$, $x \in C$.

Thus, by defn $A \subseteq C$.

defn
defn

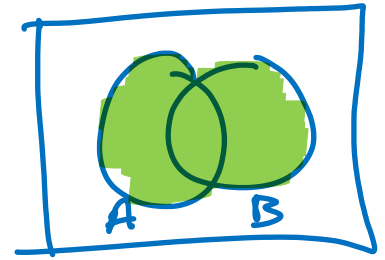
$\square$ ($\forall x$)

A, B, C
 $A \subseteq B$
 $B \subseteq C$
 $x \in A$
 $x \in B$

Operations on Sets (corresponding to $\vee, \wedge, \neg$)

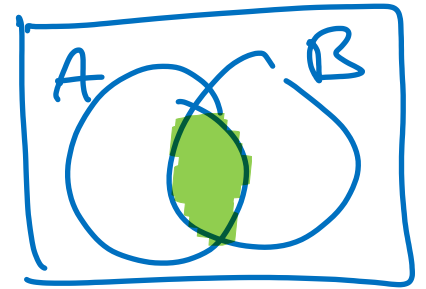
① Union: $A \cup B = \{x : x \in A \vee x \in B\}$

ex! $E = \{x \in \mathbb{Z} : x \text{ is even}\}$
 $O = \{x \in \mathbb{Z} : x \text{ is odd}\}$
 $\mathbb{Z} = E \cup O.$

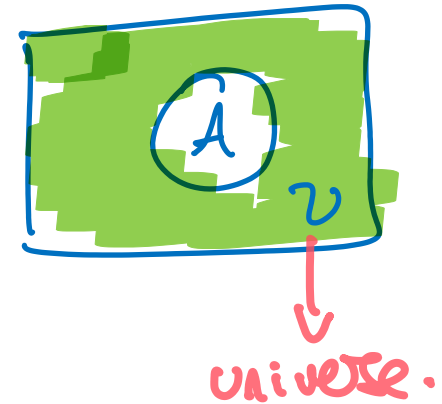


② Intersection: $A \cap B = \{x : x \in A \wedge x \in B\}$

ex! $E \cap O = \emptyset.$
 $\mathbb{Z} \cap \mathbb{R} = \mathbb{Z}$



③ Complement: $\overline{A} = \{x \in U : \neg(x \in A)\}.$
 $= U - A.$



④ Difference: $A - B = \{x \in A : x \notin B\}.$

Set Identities : Relations between sets which always hold.

ex: $\forall A, B:$

$$\overline{A \cup B} = \overline{A} \cap \overline{B}.$$
$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

} De Morgan for sets.

$$A \cup (A \cap B) = A \dots$$

"follow the logic" proofs.

Remark: To prove $A=B$, first prove $A \subseteq B$
then prove $B \subseteq A$.

Prop 2: If A, B are sets, $\overline{A \cup B} = \overline{A} \cap \overline{B}$.

Proof: Let A, B be arbitrary.

($\subseteq$) Assume $x \in \overline{A \cup B}$ is arbitrary.

By defn, $\neg(x \in A \cup B)$, which is equiv. to $\neg(x \in A \vee x \in B)$.

By de Morgan, $(x \notin A \wedge x \notin B)$

A, B
 x

By defn, $x \in \bar{A} \cap \bar{B}$. complement
By defn, $x \in \bar{A} \cap \bar{B}$, intersection as desired.

($\supseteq$)

<similar, skip>.

□

Two more Ops:

Defn: If A is a set, the power set $\leftarrow$ set of all subsets.
 $P(A) := \{ S : S \subseteq A \}$

ex: $P(\{1, 2\}) = \{ \{1\}, \{2\}, \{1, 2\}, \emptyset \}$.

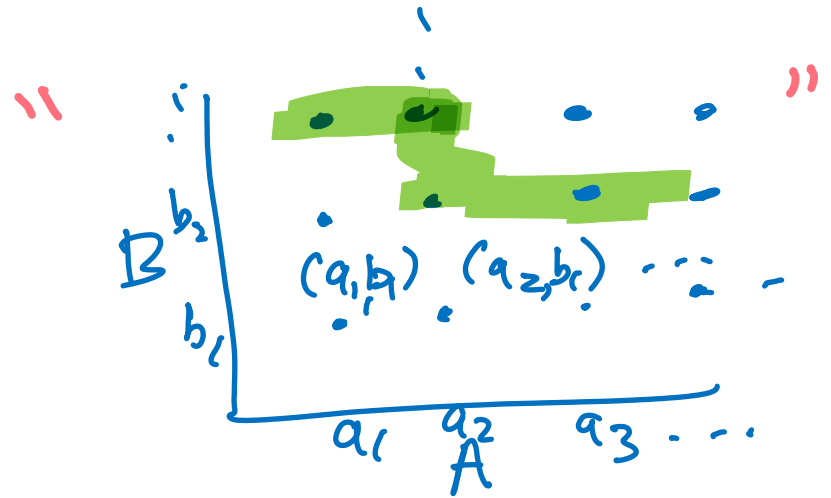
$P(\emptyset) = \{ \emptyset \}$ $\emptyset \subseteq \emptyset$

Warning: $\subseteq$ is not the same as $\in$.
 $2 \in \{1, 2\}$ $\{2\} \subseteq \{1, 2\}$.
 $2 \notin \{1, 2\}$. $\{2\} \notin \{1, 2\}$.

Be mindful of types (element vs. set).

Defn: $A \times B = \{ \underbrace{(a, b)}_{\text{ordered pair} \neq (b, a)} : a \in A \text{ and } b \in B \}$
(Cartesian Product)

Picture:



Remark: $A \times B \neq B \times A$.

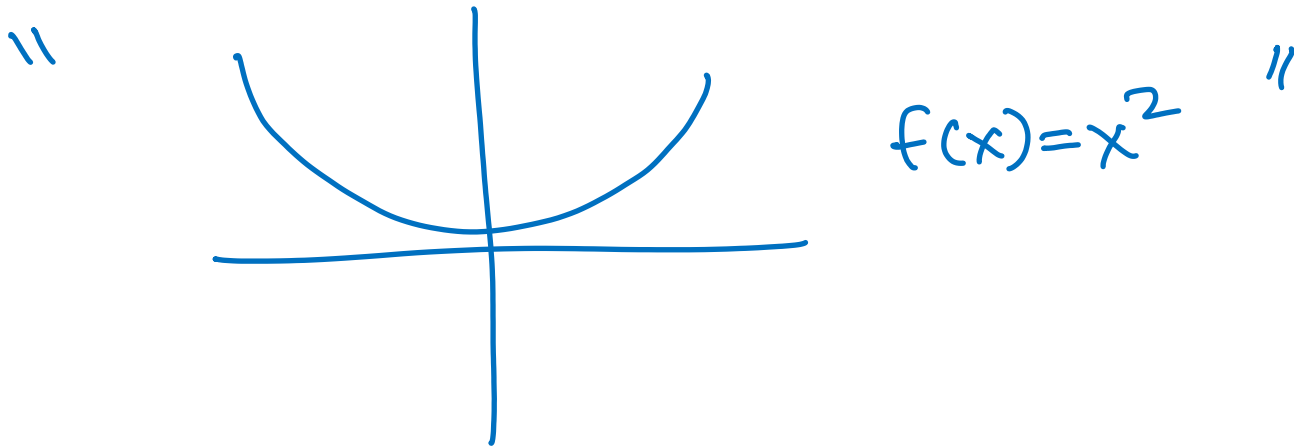
ex: $C = \{x : x \text{ is a person in class}\}$
 $S = \{s : \text{songs on spotify}\}$
 $C \times S = \{(x, s) : x \in C \text{ and } s \in S\}$
 $L = \{(x, s) \in C \times S : x \text{ has listened to } s\}$

Q) Is $\{2\} \subseteq \mathcal{P}(\{1,2\})$? No.

$\{\{2\}\} \subseteq \mathcal{P}(\{1,2\})$. Yes.

Remark: • If $C = \{c_1 \dots c_n\}$ $S = \{s_1 \dots s_m\}$
then $C \times S$ has $m \times n$ elements.
<Prove later> • If A has n elements, $\mathcal{P}(A)$
has 2^n elements.

II Functions (Revisit a familiar concept w/ more precision)



Defn:

If A, B are nonempty sets, a function

from A to B , denoted

$$f: A \rightarrow B$$

is a rule which assigns one element of B to every element of A .

The assignment is denoted

$$f(a) = b$$

" b is the image of a ."

"Given an input, produces an output."

A is called the domain.

"Set of inputs"

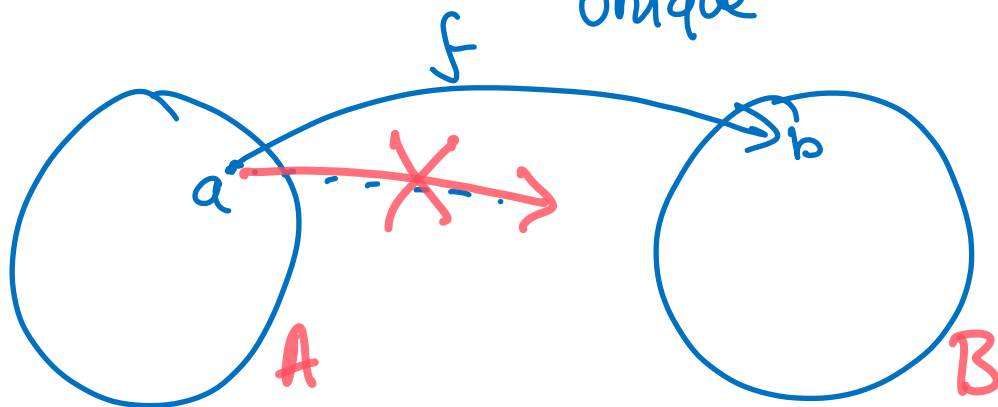
B is called the co-domain.

"Set of possible outputs."

Logically:

$$\forall a \in A \left(\exists! b \in B (f(a) = b) \right)$$

unique



ex:

A B

rule

onto

$$f: \mathbb{Z} \rightarrow \mathbb{Z}$$

$$f(x) = x^2.$$

No. -1, 2

$$g: \mathbb{R} \rightarrow \mathbb{R}$$

$$g(x) = x^2.$$

No. -1.

$$n: C \rightarrow C$$

$$n(x) = \begin{cases} \text{the person to the left of } x \\ \text{Nikhil if no such person.} \end{cases}$$

(not a formula)

No.

$$m: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$$

$$m(a, b) = ab.$$

Yes.

$$p: A \rightarrow \mathcal{P}(A)$$

$$p(a) = \{a\}.$$

$\mathcal{P}(A)$

No.

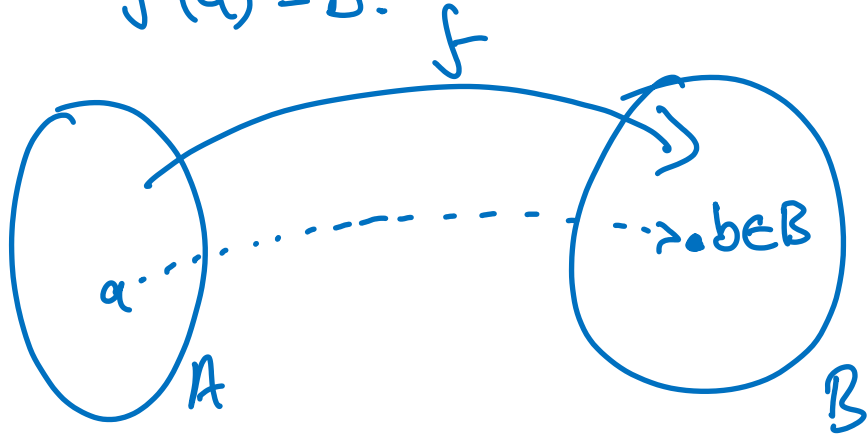
$b = \emptyset,$
 $\{a_1, a_2\}.$

$$h: \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$$

Yes.

Two Properties:

Defn: $f: A \rightarrow B$ is onto
if for every $b \in B$ there
exists an $a \in A$ such that
 $f(a) = b$.



$$\forall b \in B (\exists a \in A \ f(a) = b).$$

(depends on A, B not just f).

<Next time>.